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Article

# Investigating Errors in Alchemical Free Energy Predictions Using Random Forest Models and GaMD

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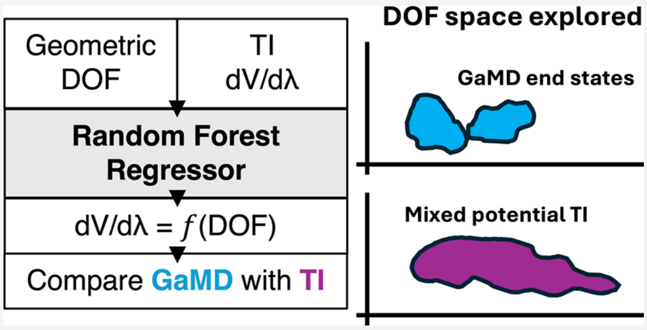
**ABSTRACT:** State-of-the-art *in silico* $\Delta\Delta G$ predictions for antibody−antigen complexes achieve an accuracy of ±1 kcal/mol. While this is sufficient for high-throughput screening or affinity maturation, it is insufficient for assessing the criticality and impact of post-translational modifications (PTMs) during clinical development. PTMs that impair binding by >50% pose a major risk to achieving the desired therapeutic bioactivity and must be controlled within defined limits to ensure product quality. A 50% loss in the dissociation constant ($K_D$) corresponds to a $\Delta\Delta G$ of +0.5 kcal/mol, thus requiring a ±0.5 kcal/mol accuracy threshold for *in silico* predictions to become practically actionable in clinical phases. In this work, we use conventional molecular dynamics thermodynamic integration (cMD-TI) to generate $\Delta\Delta G$ predictions and develop an error analysis approach using random forest (RF) models and end-state Gaussian accelerated molecular dynamics (GaMD). This approach provides untargeted insight into inadequate sampling of key degrees of freedom using only cMD-TI and end-state GaMD. We identify bulky side-chain undersampling and violation of energetically relevant interatomic interactions as major sources of error, and our GaMD-based error corrections lead to >1 kcal/mol improvements in accuracy in our most erroneous cases. When applied to a set of 13 predictions, the GaMD-based error correction reduced the root-mean-square error (RMSE) from 1.06 ± 0.22 to 0.70 ± 0.18 kcal/mol. This work introduces the application of alchemical free energy predictions to estimate PTM impacts on bioactivity and investigates the current errors that limit their practical use in clinical development.






## ■ INTRODUCTION

Antibodies are a fundamental component of the human immune system,[1] and designed therapeutic antibodies can harness the immune system to fight various diseases.[2] Therapeutic antibodies bind specifically to their cognate antigens, which either elicit antagonistic/agonistic effects or recruit immune cells for downstream effector activity depending on the mechanism of action.[3] Consequently, the binding affinity (antibody−antigen or antibody−receptor) is an essential component of antibody efficacy.

During large-scale manufacturing, therapeutic antibodies often undergo post-translational modifications (PTMs), i.e., chemical modifications such as tryptophan oxidation, aspartate isomerization, and lysine glycation.[4−7] It is imperative to patient safety and health outcomes to understand where these modifications may occur and how they may impact bioactivity, pharmacokinetics, safety, or immunogenicity.[8,9] PTMs and other attributes that impact these pharmacological factors are deemed critical quality attributes (CQAs), while those that have no impact are considered non-CQAs.[10−12] PTMs that occur on residues near the antibody−antigen (Ab−Ag) or antibody−Fc receptor (Ab-FcR) interface may impact binding affinity, which in turn may affect bioactivity, effector function, or pharmacokinetics.[5,13−15] Therefore, an essential part of CQA assessment is measuring the impact of observed PTMs on Ab−Ag and Ab−FcR binding affinity.[16,17]

Currently, assessing PTM impact on bioactivity is largely done empirically, which demands that a sufficient quantity of minor variants be enriched for SPR measurements or cell-based potency assays.[16] However, this laborious enrichment is time-consuming and not consistent across the space of antibodies and PTMs.[18,19] Furthermore, multiple PTMs often occur concurrently, which makes it impossible to analyze the impact of each PTM separately.[20,21] In contrast, PTM-modified antibody structures are more feasible to generate computationally as long as structural data for the unmodified Ab−Ag complex exists.[22−24] Thus, an accurate *in silico* tool would offer an alternative method to analyze the impact of PTMs on the Ab−Ag binding affinity and Ab−Fc interactions.

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Usually, CQA assessment is needed later in the biologics development timeline when there are relatively few candidates,[16] making it acceptable to use an expensive method if it guarantees accuracy. It is imperative for decision-makers to know whether a given PTM has a >50% impact on bioactivity, since this presents a serious risk to product quality and a control strategy must be put in place to limit the variant to a consistently low fraction of the drug product.[7] Therefore, an *in silico* method must be able to accurately detect 50% $K_D$ impact to be useful in CQA assessment, which translates to a $\Delta\Delta G$ of +0.5 kcal/mol and falls short of the state of the art in *in silico* protein−protein interface relative binding free energy (RBFE) prediction, which is around ∼1 kcal/mol.[25−30] In this article, we investigated the most significant sources of error in our thermodynamic integration (TI) predictions to eventually enable the application of *in silico* methods to CQA assessment.

Estimating the impact of a PTM near the Ab−Ag interface on binding affinity can be done by measuring the relative binding free energy (RBFE), or $\Delta\Delta G$, of the PTM on the system.[31] A large variety of approaches to computing RBFEs for protein−protein complexes and protein−ligand complexes have been published.[31−35] It is important to note that these methods for RBFE estimation have seldom been applied to CQA assessment; instead, their intended applications are for antibody design or screening.

End-point methods such as MM-GBSA[36] produce estimates not calibrated to biological scales and make approximations to increase throughput at the cost of accuracy.[37,38] Thus, although these methods have been successfully used for virtual screening[39−41] and identification of key intermolecular interactions in protein complexes,[42,43] they are less suitable for applications demanding high accuracy. Methods applied to antibody design such as machine learning-based methods have advanced rapidly in recent years,[44−47] but the chemical diversity of PTMs and lack of substantial empirical PTM binding affinity data sets create significant challenges when applying these methods to PTM $\Delta\Delta G$ estimation. Pathway methods such as umbrella sampling[48−50] compute the absolute binding free energy (ABFE) by linking binding and unbinding to a physical reaction coordinate, applying biasing potentials to force unbinding, correcting for the bias, and deriving the potential of mean force (PMF).[51] Although these approaches have demonstrated accurate predictions,[52] their accuracy relies heavily on the construction of the pathway and design of restraints to ensure convergence, which requires substantial computational cost and human expertise.[53] Indeed, they are an order of magnitude more expensive than alchemical RBFE methods.[35] Based on the level of expertise required, the expense, and the lack of any additional benefits from ABFEs versus RBFEs, we set aside umbrella sampling in favor of alchemical methods.

The state of the art for accuracy in RBFE estimation is alchemical methods,[54] e.g., free energy perturbation and thermodynamic integration (TI).[55−60] Put simply, to measure the $\Delta\Delta G$ associated with an amino acid mutation (e.g., Trp to Ala), TI exploits the thermodynamic cycle by computing the $\Delta G$ of the direct nonphysical transformation of Trp into Ala in the bound ($\Delta G_{\text{bd}}$) and unbound states ($\Delta G_{\text{ubd}}$) in a reversible way. In practice, the wild-type and mutant states are connected by the parameter $\lambda$, with $\lambda = 0$ representing the wild-type end state and $\lambda = 1$ representing the mutant end state. The $\Delta G$ of the direct nonphysical transformation is found by simulating the system at various intermediate $\lambda$ windows, collecting the derivative of the potential energy with respect to $\lambda$ (or $dV/d\lambda$) at each intermediate window, and finally computing the integral given in eq 2. Due to the thermodynamic cycle, the difference between these two quantities ($\Delta G_{\text{bd}} - \Delta G_{\text{ubd}}$) is the same as the difference between the binding free energy of the Trp complex and the Ala complex.[28,30]

Current alchemical methods are capable of predicting RBFEs of protein−protein interactions with ∼1 kcal/mol error,[25−30,61] which is still below the accuracy threshold necessary to apply these methods to CQA assessment. Previous work has identified the most likely sources of error in these calculations to be insufficient sampling,[25,62] issues with force fields,[63−65] errors in structure preparation,[66] or artifacts with mixing potentials.[67,68] Investigating and alleviating these sources of error may allow alchemical methods to attain sufficient accuracy to be practically applied to CQA assessment.

In this study, we generated a set of RBFE predictions based on experimental mutagenesis data. Since the experimental mutagenesis data sets for canonical single amino acid mutations are much richer than PTMs, we primarily focused on canonical amino acid mutation data for structurally determined systems, such as mab1/mab2/mab3 from Gao et al.[69] and hu4D5−5 from Kelley and O'Connell.[70] This allowed us to temporarily leave aside the additional confounding variable of developing new force fields for noncanonical amino acids.

We investigated the root causes of the highest error cases in our predictions using random forest (RF) model[71] impurity-based feature importance to identify influential degrees of freedom (DOF) on $dV/d\lambda$. We compared the sampling of these influential DOF with free energy profiles derived from enhanced sampling at the end states via Gaussian accelerated molecular dynamics (GaMD) to verify the proper sampling of the important DOF in TI $\lambda$ production simulations. This approach revealed that bulky side-chain undersampling and violation of energetically relevant interatomic interactions were the culprits for our most inaccurate predictions. We explored methods to alleviate these sources of error and enable the eventual application of these methods to *in silico* CQA analysis.

## ■ METHODS

**Empirical Data.** Experimental data was collected from Kelley and O'Connell,[70] Kelley et al.,[72] and Gao et al.[69] Table S1 contains the published empirical value for each point mutation. Some hu4D5−5 mutants had multiple published values, and we corresponded with Kelley to clarify which was the most reliable estimate. Equation 1 was used to convert $K_D$ values into $\Delta\Delta G$ kcal/mol values.

$$\Delta\Delta G = RT\ln(K_{\text{D1}}/K_{\text{D2}}) \tag{1}$$

The same was done for data in Gao et al.,[69] i.e., a 5-fold change in binding affinity by SPR is equal to a $K_D$ ratio of 0.2, which was plugged into the same equation.

**Preparation of Structures.** For hu4D5−8, we downloaded a cryo-EM structure (PDB: 6OGE) from the PDB.[73] This was the only available structure in the PDB with the full Erbb2 antigen bound to the hu4D5 Fab that did not miss significant regions in the antigen loop near the interface.

For hu4D5−5, we took the hu4D5−8 cryo-EM structure and made modifications to VH-V102Y and VL-E55Y to convert the structure to hu4D5−5. To confirm that there were no conformational shifts due to these mutations, we aligned a crystal structure of the Fab of hu4D5−4[74] (PDB: 1FVD) with the cryo-EM structure of hu4D5−8, since hu4D5−4 was quite similar to hu5D5−5, except for VH-A78L. During simulations with hu4D5−8 and hu4D5−5, we truncated the Erbb2 antigen and left only 135 residues on the C terminus to save computational expense, since the rest of the antigen was far (>40 Å) from the Ab−Ag interface. The backbone atoms of the first 13 residues following the truncation were restrained with a weight of 1.0 kcal/(mol·Å$^2$) throughout all simulations of the bound complex. For mab1, mab2, and mab3 in Gao et al.,[69] we used crystal structures from the PDB.

For the structures of the mutants, we kept the backbone atoms and mutated the side chains. We used tLEaP[75] and parmed[76] tiMerge within AmberTools 20[77] to prepare the topologies for the simulations. tLEaP[75] was used to solvate the structure with a 10 Å radius waterbox and to add neutralizing ions as well as $Na^+$ and $Cl^-$ ions to create a 0.15 M NaCl environment. All simulations were done in AMBER20[77] with force field ff14SB[78] and water model TIP3P.[79] For force field parameterization of methionine sulfoxide (necessary for estimating the impact of methionine oxidation), we used Gaussian 09[80] with the HF/6−31G* basis set to generate restrained electrostatic potential (RESP) partial charges and antechamber[81,82] with the Generalized Amber Force Field[83] (GAFF) to assign atom types and bonded parameters.

**TI Theory.** Alchemical transformations were used to estimate the impact of canonical amino acid mutations and PTMs on the Ab−Ag binding affinity. TI exploits the thermodynamic cycle (Figure 2, top left box) and allows the $\Delta\Delta G$ to be computed using a nonphysical, reversible transformation from the wild-type to the mutant residue in both the bound and unbound configurations ($\Delta\Delta G = \Delta G_{\mathrm{bd}} - \Delta G_{\mathrm{ubd}}$). To estimate the $\Delta G$ of a nonphysical transformation using TI, one must simulate the system at various intermediate $\lambda$ windows and collect the derivative with respect to $\lambda$, or $\mathrm{d}V/\mathrm{d}\lambda$, at each intermediate value. The integral in eq 2 gives the $\Delta G$ for a transformation in a given system; in this study, the $\lambda$ windows were set to a 12-point Gaussian quadrature to numerically approximate the integral into a weighted sum as given in eq 3. Subtracting the $\Delta G$ of the unbound state transformation from the bound state transformation yields the total $\Delta\Delta G$, or the change in the binding free energy of the Ab−Ag complex upon a mutation or PTM.

A more comprehensive treatment of the theory underpinning alchemical methods, which is beyond the scope of the paper, can be found elsewhere.[51,84]

$$\Delta G = \int_{\lambda=0}^{\lambda=1} \frac{\mathrm{d}V}{\mathrm{d}\lambda}\mathrm{d}\lambda \tag{2}$$

$$\Delta G \approx \sum_{j=1}^{12} w_j \cdot \left\langle \frac{\mathrm{d}V}{\mathrm{d}\lambda} \right\rangle_{\lambda_j} \tag{3}$$

**TI Protocols.** *Preliminary Structure Relaxation.* For all neutral (non charge-changing) mutations, the structures underwent 5000 cycles of minimization (10 cycles of steepest descent, the rest conjugate gradient) and then 0.8 ns of constant volume heating from 10 to 300 K, with a Langevin thermostat and collision frequency gamma = 5.0 1/ps. Then, they underwent 4 ns of constant pressure relaxation with gamma = 1.0 1/ps and finally 5 ns of constant volume relaxation. The first three steps were done with backbone restraints with a weight of 10.0 kcal/(mol·Å$^2$) for minimization and 1.0 kcal/(mol·Å$^2$) for heating and constant pressure relaxation and were removed for the constant volume relaxation step. All four steps were done at $\lambda = 0.5$, and a time step of 1 fs was used in all simulations.

*λ Equilibration.* The $\lambda$ equilibration order was adapted from He et al.,[85] and equilibration was done with collision frequency 1.0 1/ps at constant volume for 1 ns. The $\lambda = 0.5$ structure from the previous section was used as the initial structure for $\lambda = 0.4378$ and $\lambda = 0.56262$ equilibration, and then the final structure after 1 ns of equilibration for these two $\lambda$ values was used as the initial structure for their neighbors to the left and to the right, respectively. This bidirectional serial equilibration allowed us to gently equilibrate structures at $\lambda$ values from $\lambda = 0$ to $\lambda = 1$.

*Production.* Production simulations were run for 5 ns for each of the 12 $\lambda$ windows. Simulations were constant volume, and a Langevin thermostat was used, now with gamma = 2.0 1/ps. A 12-point Gaussian quadrature was used to evaluate the $\mathrm{d}V/\mathrm{d}\lambda$ integrals. Five replicates were done for both the bound and unbound states, and the difference of the average of the bound and unbound states was the final computed $\Delta\Delta G$ of the mutation. Statistical uncertainty was calculated by propagation of error (summing the square of the bound and unbound standard errors and taking the square root of the sum).

*Two-Step/Charge-Changing TI Protocol.* We used the two-step approach for all charge-changing mutations,[86] i.e., transformation was separated into a van der Waals and charging step. For example, the mutation E55Y would be separated into a charging step (E55 to uncharged E55) and a van der Waals step (uncharged E55 to Y55). The uncharged end states were neutralized with Amber's crgmask command. The $\Delta\Delta G$ of the transformation was found by taking the sum of the $\Delta\Delta G$ of both parts, and the standard error was calculated by propagation of error, as detailed in the previous section.

For the van der Waals step, everything was done as described above with the crgmask for the charged residue. For the charging step, we used a coalchemical neutral ion correction to resolve the net charge in the simulation box as described in previous literature with the softcore atoms mapped to avoid the particle collapse problem and the coalchemical ion mapped to its duplicate.[66,87,88]

*One-Step/Charge-Changing TI Protocol.* In addition to the two-step protocol, we also used a one-step protocol for mab2 VL-Y → R and hu4D5−5 VH-R50A to ensure that any errors in the two-step protocol were not an artifact of the protocol itself. We used a longer relaxation protocol, identical to the one described below in the GaMD Simulations section, the same coalchemical ion neutral correction as in the two-step protocol, and the second-order smoothstep softcore potentials[68] with their recommended parameters (gti_lam_sch = 1, gti_ele_sc = 1, gti_vdw_sc = 1, scalpha = 0.2, scbeta = 50).

*Distance-Based Restraints.* In order to correct for disrupted salt bridges during mab2 VL-Y → R and hu4D5−5 VH-R50A, distance-based restraints were used to restrain the distance between the atoms participating in the relevant salt bridges (nmropt = 1 in Amber 20). The distance range ($r_1$, $r_2$, $r_3$, $r_4$) for the restraints was determined by the GaMD free energy profile, and the restraints had a force constant of 10.0 kcal/(mol·Å$^2$).

**GaMD Protocols.** *GaMD Simulations.* For all end states, 5 replicates of 300 ns GaMD[89] were run, using a longer relaxation protocol, adapted from Carlos Simmerling's


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relaxation of explicit water systems tutorial.[90] The initial minimization was for 1000 cycles (20 steepest descent, the rest conjugate gradient) with all solute atoms restrained with a weight of 100 kcal/(mol·Å$^2$). Then, 1 ns heating from 100 to 300 K and 4 ns constant pressure relaxation with the same restraints were done before reducing restraints to weigh 10 kcal/(mol·Å$^2$) for the second relaxation step (1 ns). Then, a second minimization step (5000 cycles, 30 steepest descent, the rest conjugate gradient) was carried out, now with only backbone atoms restrained at weight 10 kcal/(mol·Å$^2$). After this, 4 consecutive 1 ns constant pressure relaxation steps reduced restraints from 10 to 1 to 0.1 to 0, and then finally, a 4 ns unrestrained constant volume relaxation was done. A Langevin thermostat was used as above, with a collision frequency of 1.0 1/ps for all steps except heating (5.0 1/ps).

GaMD was done at constant volume, with 2 fs timesteps. Tolerance was set to 0.000001 (1/10 of default), with Langevin frequency gamma = 2.0 1/ps, sigma0P = 6.0, and sigma0D = 6.0. The system ran for 2 ns cMD initially, with potentials collected from 0.4 ns onward, and then 6 ns of biasing MD, with potentials being updated from $t = 2.4$ to $t = 6$ ns. Threshold energy was set to lower bound $V = E_{max}$.

To generate 1-D and 2-D reweighted pmf plots of any given metric, the metric was measured for all frames of the GaMD simulation, and PyReweighting-1D.py or PyReweighting-2D.py[91] was used to produce a pmf value using second-order cumulant expansion. This pmf value was converted to a free energy profile by taking the inverse exponential $e^{-pmf}$, which was integrated to find the area and volume of a given state. Convergence of free energy profiles was assessed by reweighting at various fractions of the full simulation length and measuring cosine similarity or relative area of multiple microstates over time.

*GaMD Trajectory Clustering.* CPPTRAJ[92] was used for clustering. Due to memory and time constraints on the computing cluster, every 10th frame was taken from each trajectory. All trajectories were loaded and stripped of solvent molecules and ions, and then the rms command was used to align all frames to the first frame, using only the RMSD of atoms within 6 Å of the softcore residue in the first frame (e.g., the command was rms first:448<:6.0&!@H= if the residue being transformed was residue 448). Then, hierarchical agglomerative clustering with average linkage, epsilon 3.0, and 100 total clusters was used to generate clusters. The same RMSD mask used for alignment was then used for clustering, and the sieve was set to 100. Reweighting by second-order cumulant expansion was done to find the lowest energy clusters from the GaMD trajectories. In the hu4D5−5 VL-N53 end-state GaMD trajectory, we observed unbinding of the Ab−Ag complex in two of the five replicates and therefore excluded them from clustering.

**RF Error Analysis/Correction.** *Finding Influential Features.* All frames belonging to the lowest energy GaMD cluster were extracted from the GaMD trajectories, and then the VMD HBonds[93] plugin was used to find any interatomic distances that occurred with nontrivial (>1%) frequency in the lowest energy GaMD cluster trajectory. These interatomic distances, along with rotamers within 5 Å of the softcore region in the cluster centroid, were included as input features into the random forest model. For our dependent variable, we did not want to simply use d$V$/d$\lambda$ since the derivative for each $\lambda$ is weighted differently in the Gaussian quadrature sum. We transformed this variable into the product of the Gaussian quadrature weight for the particular $\lambda$ value and d$V$/d$\lambda$.

With all possible features collected, supervised learning was used to identify an underlying relationship between the geometric properties measured during TI (softcore or near-softcore rotamers and interatomic distances) and the energetic properties (Gaussian quadrature weighted d$V$/d$\lambda$ values), both of which were computed at every frame of the TI $\lambda$ production trajectory. Models were trained separately for each mutation, and machine learning calculations were done in scikit-learn.[94]

Cross correlation of the feature set was monitored, and features with a Pearson correlation coefficient $r > 0.5$ were removed; this is a common feature selection step to reduce feature redundancy, minimize overfitting, and improve model interpretability. Features were then scaled, and recursive feature elimination with an underlying decision tree regressor with maximum depth 5 was used to select the top 75% of the features, eliminating 5% of the features with each iteration. Then, these features were used to train a RF regression model with maximum depth 10, containing 200 total trees, using a maximum of 60% of the features provided by the earlier step, with internal nodes needing a minimum of 14 samples and leaf nodes needing a minimum of 7 samples; these hyper-parameters were found using Bayesian hyperparameter tuning. The hyperparameter search was configured to run for 200 iterations, with 50 points initially generated through Latin hypercube sampling, followed by 150 Bayesian-guided iterations. Since the goal of the model was to use geometric DOF (i.e., the independent feature set) to explain as much of the variance in TI energetics as possible, we used mean training and test $R^2$ after 5-fold cross validation as our objective function to guide the hyperparameter search and chose the set given above to maximize model $R^2$ for each mutation.

Repeated stratified 5-fold cross validation (repeated 5 times, stratified by $\lambda$ value) was used to get a test $R^2$, and finally the mean impurity-based feature importances (using scikit-learn's default feature_importances_attribute) from the 25 different RF models during 5 × 5-fold cross validation were used to identify the most energetically relevant features.

After the RF model training and cross-validation, if the model had a mean test $R^2 > 0.4$, then the distribution of the most relevant features during TI was compared with their GaMD end-state free energy profile. This cutoff was chosen since the independent and dependent feature sets were highly noisy, and we wanted to ensure a strong signal between geometric features and energetic d$V$/d$\lambda$ values and avoid cases with spurious correlations.

The TI data was corrected by filtering to only the TI frames that conformed to the free energy profile. If distinct predominant microstates were identified in the free energy profile, the $\Delta G$ of those microstates was separately computed and added by linear combination and weighted by the area of the microstate in the free energy profile. If a particular $\lambda$ in a microstate had less than 100 frames (5 runs, 200 frames each, means the original runs had 1000 frames per $\lambda$), more TI data was collected to ensure robust sampling of all $\lambda$ windows in the microstate. In a few microstates, even with extensive TI sampling, there remained insufficiently sampled $\lambda$ values. This was due to weakened potentials (e.g., salt bridge violation; see Discussion for more details) or high potential energy barriers restricting comprehensive sampling of all $\lambda$s at all microstates during TI. In such cases, an approximate microstate $\Delta\Delta G$ was


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computed by concatenating the insufficient d$V$/d$\lambda$ values with those of the other microstates.

**Computing Uncertainty Metrics for Corrections with Bootstrapping.** We could not use standard error to estimate the uncertainty of our corrected $\Delta\Delta G$ predictions as we did in our initial predictions, since we aggregated the TI $\lambda$ sampling across all replicates to compute $\Delta G$s for a given microstate. Thus, we used bootstrapping to compute uncertainty metrics for each RF + GaMD-corrected $\Delta\Delta G$ prediction. We sampled each of the $\lambda$ values 200 times with replacement and computed the $\Delta\Delta G$, repeating this process 1000 times to produce a distribution of bootstrap-sampled corrected $\Delta\Delta G$ values. We chose to sample 200 times from each $\lambda$ value since our initial estimations were computed with 200 frames per $\lambda$ value. The final values and their uncertainties (95% confidence intervals) were calculated by taking the mean, 2.5th percentile, and 97.5th percentile, respectively, from the bootstrap-sampled distribution. In principle, due to the central limit theorem, the greater and lesser confidence intervals should be approximately equivalent in magnitude. We verified that this applies to our data and thus concisely report measurements with their uncertainties in this article as (**mean ± uncertainty**), with the given uncertainty value being the maximum of the two confidence intervals. For example, a distribution with mean 0 and confidence interval [−0.99, 0.98] would be given as "0 ± 0.99".

In order to provide a consistent comparison of the initial and corrected $\Delta\Delta G$ estimates, we used the same method to compute 95% confidence intervals for our initial uncorrected $\Delta\Delta G$ predictions as well; this gave us a second uncertainty metric for our initial estimates along with the standard error of the 5 biological replicates. Using this set of means and 95% confidence intervals for both our initial and corrected predictions, we produced a distribution of 1000 root-mean-squared errors (RMSE) and mean absolute errors (MAE) for the data set by iterating through each prediction, taking a random sample based on the mean and 95% confidence interval, calculating RMSE/MAE from these random samples, and repeating this process 1000 times. The 95% confidence intervals for the RMSE and MAE were obtained directly from these distributions, which allowed us to measure prediction accuracy while also accounting for highly imprecise predictions.

## ■ RESULTS

**Benchmarking TI Workflow with a Mutagenesis Data Set.** We benchmarked our TI workflow on quantitative and qualitative empirically determined binding data from three antibody systems: hu4D5−5, mab1, and mab2. Our empirical data set has a bias toward mutations toward alanine, a standard problem in data set generation for protein−protein interface RBFEs. Using the protocol established with canonical amino acids, we estimated the impact of methionine oxidation on three labile residues in mab3 (Table S3).

Our TI workflow featured a standard one-step alchemical transformation (12 $\lambda$ windows with softcore potentials) for non charge-changing mutations and a two-step alchemical transformation (one van der Waals step supplemented with a decharging or charging step) with coalchemical neutral ion transformation for charge-changing mutations. Where quantitative empirical results were available ($n$ = 20), the RMSE between empirical $\Delta\Delta G$ and our *in silico* TI $\Delta\Delta G$ results was 0.94 kcal/mol (Figure 1). A comprehensive table containing the empirical and predicted values in our data set is given in Table S3.

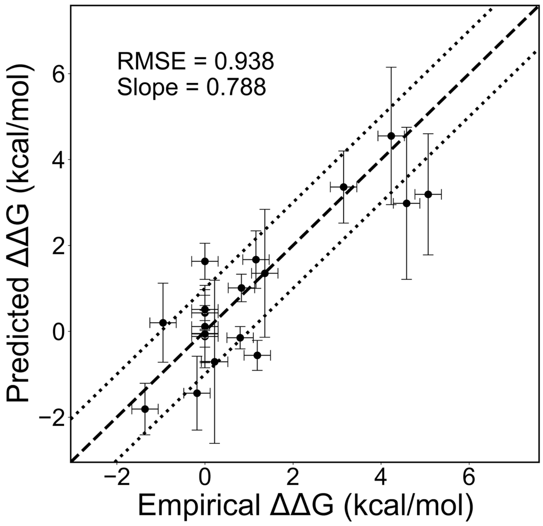


**Figure 1.** Empirical vs predicted $\Delta\Delta G$ plot, containing all cases with quantitative empirical results—the slope of the line of best fit is 0.788 and RMSE is 0.94. Vertical error bars are standard error from 5 biological replicates per case, and horizontal error bars are ±0.3 kcal/mol (inherent SPR method variability). The dashed line is $y = x$, and the dotted lines bound the ±1 kcal/mol error region.

Stratifying our initial results by systems involving a bulky side-chain residue (e.g., Phe, Tyr, or Trp) or a charge-changing transformation revealed an accuracy disparity. Larger inaccuracies were observed in the charge-changing mutations ($n$ = 4, RMSE = 1.15 kcal/mol) and the bulky side-chain mutations ($n$ = 8, RMSE = 1.18 kcal/mol), while transformations involving neither charge-changing nor bulky side chains yielded the best accuracy ($n$ = 9, RMSE = 0.55 kcal/mol). Among the qualitative empirical data, the system with the largest discrepancy between empirical and *in silico* results (mab2 Y → R) happened to be both a charge-changing and bulky side-chain transformation. We focused on identifying the root causes of TI prediction inaccuracies in our top qualitative error case (mab2 Y → R) and our top four quantitative error cases (hu4D5−5 R50A/F53N/W95A and mab2 T → Y).

**Winnowing for Influential DOF Using the RF Model.** In order to effectively interrogate, diagnose, and ultimately mitigate the root causes of TI prediction inaccuracies, we needed a means of winnowing for energetically relevant DOF among the noisy plethora of irrelevant DOF. We achieved this by training an RF model on d$V$/d$\lambda$ using the DOF localized to a 5 Å radius of the softcore region as inputs. We rationalized that the DOF closest to the softcore regions had the highest likelihood of influencing d$V$/d$\lambda$ (Figure 2). The purpose of the model was to identify the DOF that were most strongly correlated with TI d$V$/d$\lambda$ using the RF model impurity-based feature importance. For the most erroneous cases, all trained models had $R^2$ > 0.4, and the greatest correlates to d$V$/d$\lambda$ for each respective model are listed in Table 1.

We tested whether the most energetically influential DOFs were adequately sampled during TI by contrasting their TI $\lambda$ production statistics with DOF free energy profiles derived from end-state GaMD. This was based on the assumption that end-state GaMD simulations are accelerated simulations and are thus more likely to be converged. We discovered


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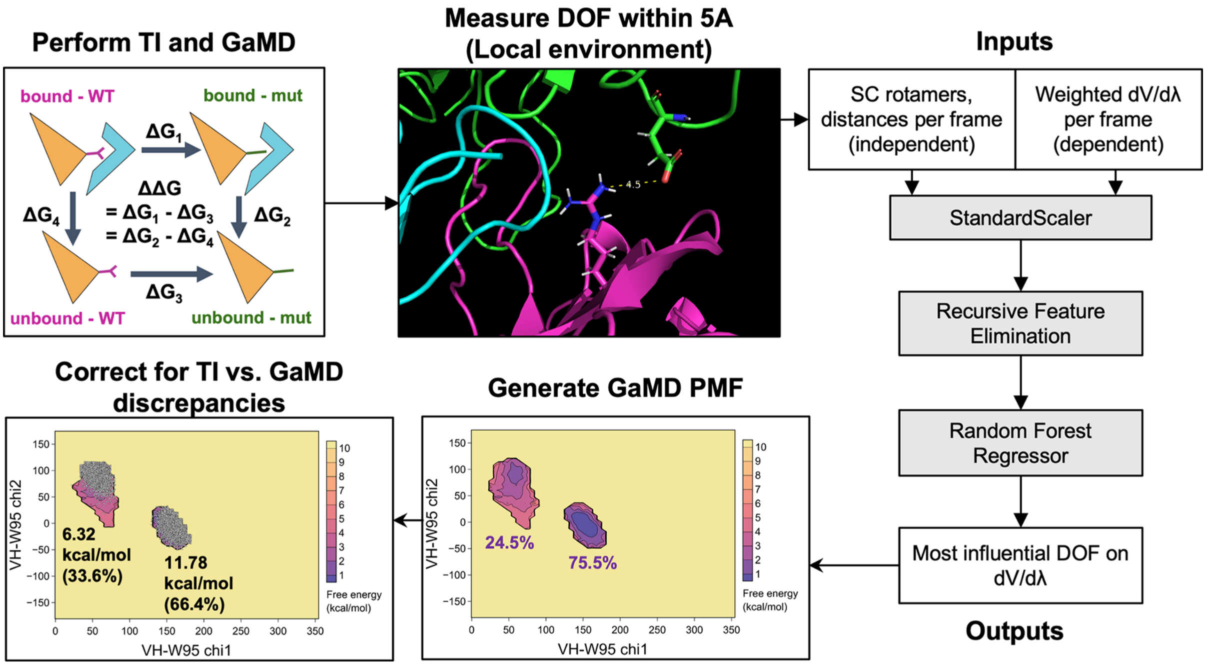


**Figure 2.** Graphical diagram of the TI calculation + error mode analysis approach. The top left displays initial cMD-TI; the top middle displays the residue to be mutated (magenta sticks), a nearby side-chain rotamer (green sticks), and a key interatomic distance (yellow dotted line); the far right displays the RF model; the bottom middle displays the end-state GaMD free energy profile of the most influential DOF; and the bottom left displays a comparison of TI $\lambda$ production sampling with the end-state GaMD free energy profile.

**Table 1. Top 5 Most Influential DOF for the Most Erroneous Cases, Identified by the Random Forest Model**[a]

| rank | hu4D5−5 VH-W95A | hu4D5−5 VH-R50A (charging step) | mab2 VL-Y → R (charging step) | mab2 VH-T → Y | hu4D5−5 VL-F53N |
|---|---|---|---|---|---|
| 1 | **VH-W95 chi1** | **Ag-E71:VH-R50 salt bridge dist** | **Ag-D161:VL-R49 salt bridge dist** | **Ag−V117:VH-Y53 H-bond dist** | **Ag−C117:VL-N53 H-bond dist** |
| 2 | **VH-W95 chi2** | VH-R50 chi1 | VL-R49:VL-S50 H-bond dist | VL-Y53 chi1 | VL-N53 chi1 |
| 3 | VH-R94 chi4 | VH-V48 chi1 | VL-S53 chi1 | VL-T53 chi1 | Ag-M102 chi1 |
| 4 | VH-F100 chi1 | VL-T94:VH-R50 H-bond dist | VL-S50 chi1 | VL-T53:VL-N51 H-bond dist | Ag−N53 chi2 |
| 5 | Ag-E71 chi3 | Ag-E71 chi3 | Ag-R157 chi1 | VL-Y53 chi2 | Ag−N120 chi1 |

[a]Bolded features were examined for sampling issues using GaMD free energy profiles. Feature importance values are in Table S4; DOF definitions are in Supporting Information Section 4.2.

inadequate or inaccurate sampling of energetically relevant DOF in each of our five most inaccurate cases and hypothesized that correcting for these sampling discrepancies between TI and GaMD would meaningfully improve $\Delta\Delta G$ prediction accuracy.

**Undersampling Impacts $\Delta\Delta G$ Prediction for Bulky Side-Chain Transformations (hu4D5−5 VH-W95A).** The most inaccurate prediction was for hu4D5−5 VH-W95A, which had a 1.88 kcal/mol error. The RF model found that the VH-W95 side-chain rotamers had the most influence on TI $dV/d\lambda$. Comparing the 2-D GaMD free energy profile for this side-chain dihedral pair to the conformational space explored during TI revealed that the TI W95 rotamer distributions did not conform to the GaMD free energy landscape (Figure 3). Moreover, the relative sampling of the two predominant microstates during TI did not match the relative areas of the same microstates, according to the GaMD free energy profile. This had energetic implications since the two predominant microstates had significantly different $\Delta G$ values in both the bound and unbound state calculations (Figure 3). Based on this observation, we devised a naïve correction approach to mitigate this source of error by filtering out any TI frames in which the W95 rotamer pairings did not belong to either microstate and computed the $\Delta G$ of both microstates separately. Since there were $\lambda$ values in both microstates that lacked sufficient sampling, we augmented our TI $\lambda$ production data with two additional TI simulations.

The GaMD-corrected overall $\Delta G$ for the bound and unbound states was determined by computing a linear combination of the two microstate $\Delta G$s, weighted by the relative microstate areas observed from the GaMD free energy profile (Figure 3). While the uncorrected $\Delta\Delta G$ of the augmented simulations remained quite inaccurate (1.88 kcal/mol error), our correction approach with sufficient sampling of each microstate yielded an improved prediction error of 0.44 kcal/mol.

**Loss of Salt Bridge Impacts $\Delta\Delta G$ Prediction for Select Charge-Changing Mutations (hu4D5−5 VH-R50A and mab2 VL-Y → R).** When we applied our error analysis approach to both the van der Waals and charging steps for charge-changing mutations hu4D5−5 VH-R50A and mab2 Y → R, we found that salt bridges had a major influence on charging step $dV/d\lambda$ for both cases. Corrections to the van der Waals steps made a trivial impact in both cases (Table S2).


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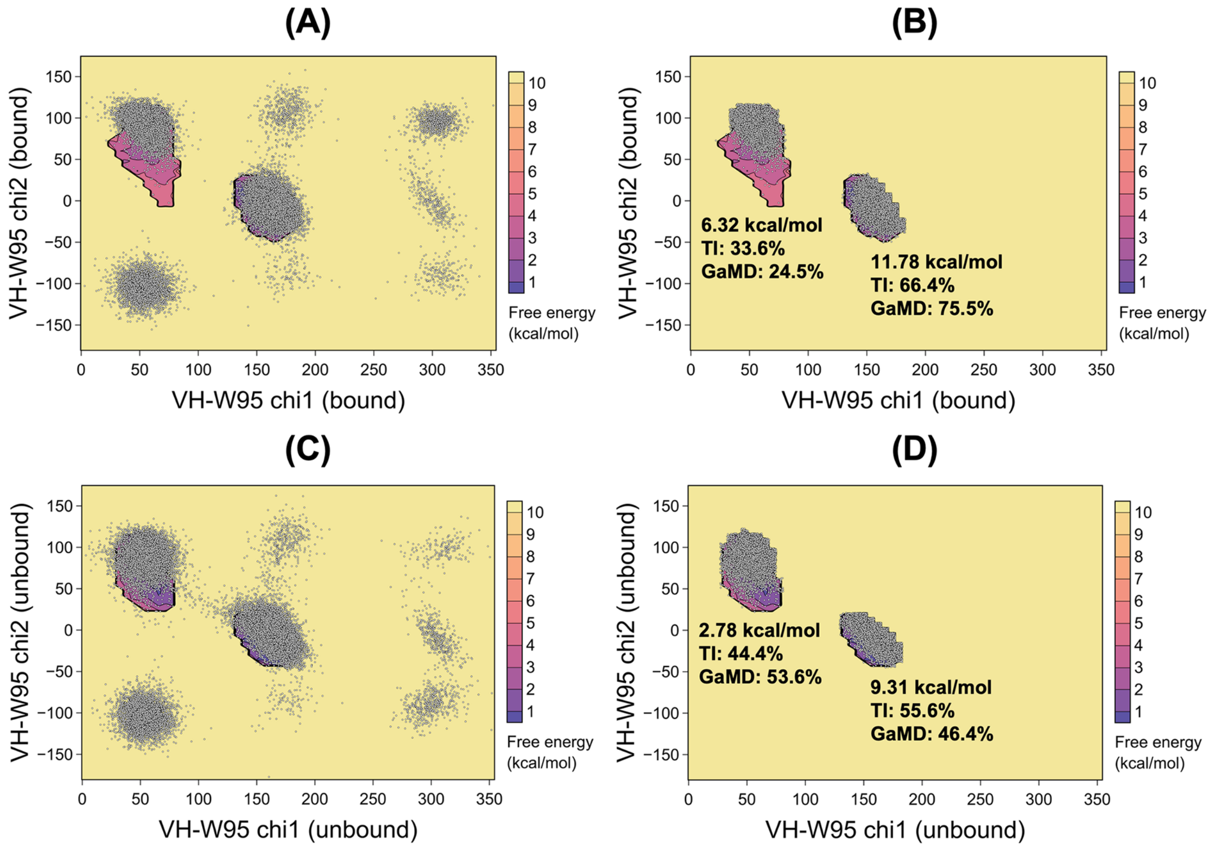


**Figure 3.** Comparison of the GaMD 2-D free energy profile of VH-W95 chi1/chi2 rotamer space (heatmap) and chi1/chi2 rotamer combinations explored during each frame of TI (scatter plot). (A) is the full bound state TI $\lambda$ production data overlaid onto the bound state GaMD free energy profile for the W95 chi1/chi2 rotamers. (B) is the same plot as (A) only with the TI $\lambda$ production sampling filtered to conform to the GaMD free energy profile. (C) is the full unbound state TI $\lambda$ production data overlaid onto the unbound state GaMD free energy profile for the W95 chi1/chi2 rotamers. (D) is the same plot as (C) only with the TI $\lambda$ production sampling filtered to conform to the GaMD free energy profile.

For the hu4D5−5 VH-R50A charging step, the RF model found that the distance between VH-R50 and Ag-E71 had the most influence on TI d$V$/d$\lambda$. The GaMD free energy profile for this distance (Figure 4) suggested that this interaction was a salt bridge (<5 Å), which is visualized in Figure 5A. When we plotted a histogram of the VH-R50:Ag-E71 distance during the charging step of the TI $\lambda$ production frames, we observed that the distance exceeded 5 Å for a majority of the TI frames (Figure 4A).

Similarly, for the mab2 VL-Y → R charging step, the RF model found that the distance between VL-R49 and Ag-D161 was the most influential degree of freedom. The GaMD free energy profile indicated that this interaction was a salt bridge (visualized in Figure 5B), and a majority of the TI frames did not conform to the free energy profile (Figure 4D). This is concerning because the charged topology should theoretically retain the salt bridge throughout all $\lambda$ windows in order to fully estimate the impact of its breaking upon a perturbation. We hypothesized that the violation of these salt bridge interactions during the TI $\lambda$ production simulation was the culprit for our $\Delta\Delta G$ error in both cases.

To test this hypothesis, we recalculated the charging step bound state $\Delta G$ using only the frames that contained the salt bridge (Figure 4B,E). Both resulting charging step $\Delta G$ adjustments improved the total $\Delta\Delta G$ prediction by >1 kcal/mol, while corrections for discrepancies in sampling during the van der Waals step made negligible changes. However, this filtering approach dramatically reduced the sample size (by 89% for hu4D5−5 VH-R50A and 57% for mab2 VL-Y → R) since the key salt bridges were not present in many frames of the charging step, as mentioned above.

To achieve a more statistically robust correction and further test our hypothesis that the disruption of key salt bridges during TI was a significant source of error, we used NMR restraints as an orthogonal means of enforcing the integrity of key salt bridges during the charging steps (Figure 4C,F). The restraints were applied to the charged and uncharged topologies during the bound state charging step and improved the accuracy by >1 kcal/mol (Table 2). Note that we neglect to account for the free energy of restraining the side chain. However, we believe that the magnitude of the error introduced by neglecting this is significantly eclipsed by the error introduced by the salt bridge violation.

The previous results were found using a two-step protocol for charge-changing modifications, but a one-step transformation using smoothstep softcore potentials is similarly susceptible to the aforementioned salt bridge violation during TI (Figures S1 and S3). The NMR restraint correction approach also effectively reduced the $\Delta\Delta G$ error in the one-step transformations by enforcing the salt bridge interaction (Table S5).

**Key Intermolecular Hydrogen Bonds Are Not Properly Captured during TI (mab2 VH-T → Y, hu4D5−5 VL-F53N).** When we applied our model to mab2 VH-T → Y and hu4D5−5 VL-F53N, we found that the most influential features on d$V$/d$\lambda$ were the distance between the VH-Y53 and


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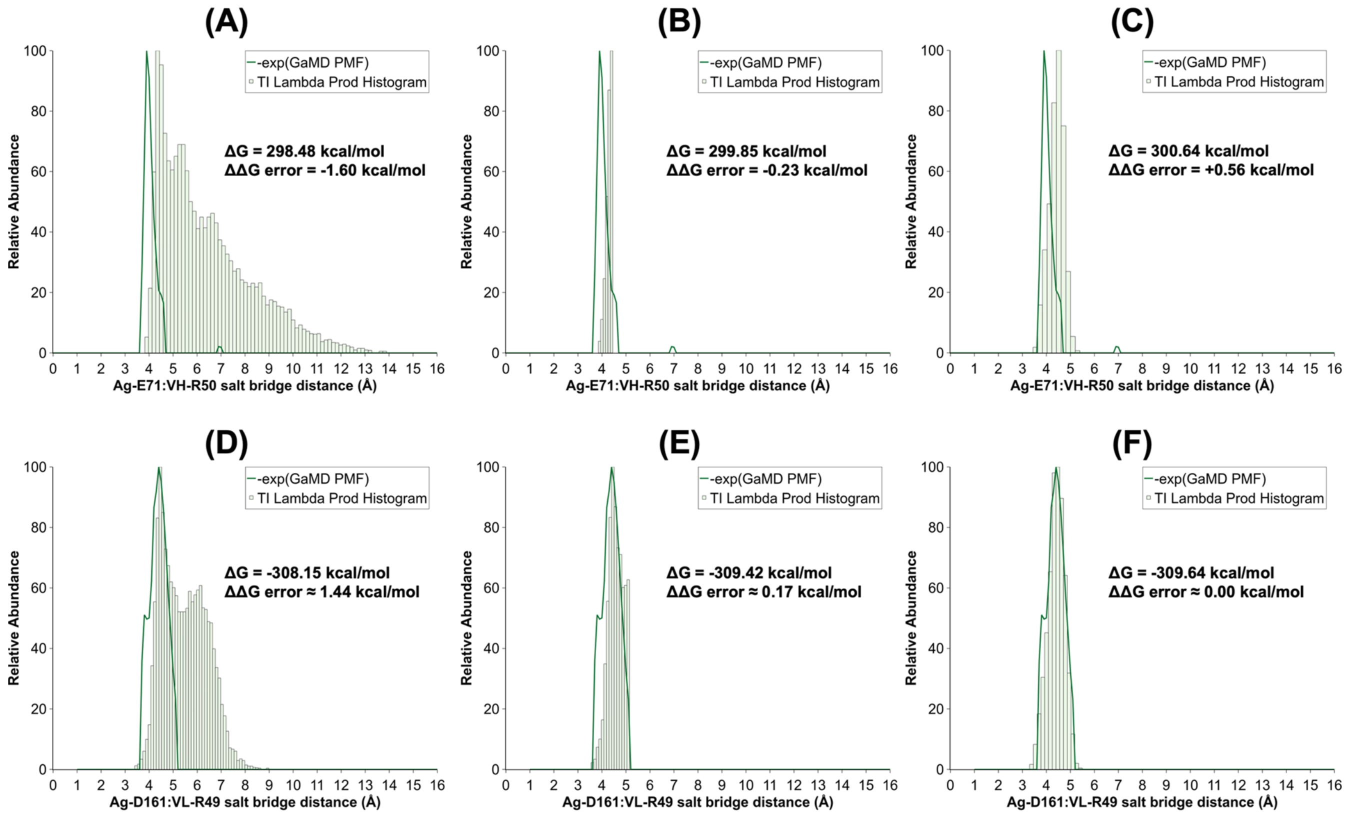


**Figure 4.** (A) is the hu4D5−5 VH-R50A bound state charging step TI $\lambda$ production data (histogram) overlaid onto the 1-D GaMD free energy profile of the key Ag-E71:VH-R50 salt bridge (line graph). (B) is the same plot as (A) but with the TI data filtered to include only the frames that included the salt bridge, i.e., distance < 5 Å. (C) is a new VH-R50A bound state charging step simulation with NMR restraints to constrain the salt bridge. (D−F) are the same but with mab2 VL-Y → R.

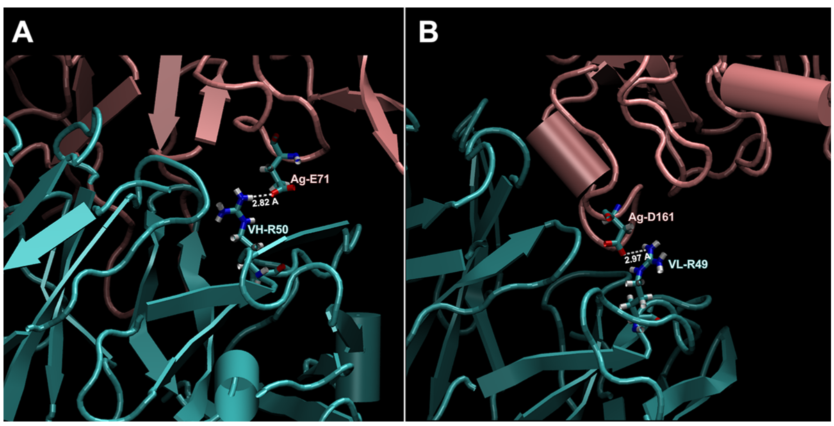


**Figure 5.** (A) Visualization of the Ag-E71:VH-R50A salt bridge between Ab/Ag of the hu4D5−5 complex. (B) Visualization of Ag-D161:VL-Y → R between Ab/Ag of the mab2 complex. Photographs taken in VMD.

Ag−V117 side chains and the distance between the VL-N53 side chain and the Ag−C117 backbone, respectively. When comparing the distribution of these distances during TI to the GaMD free energy profile, we again found that these key interatomic interactions were absent from a significant subset of TI frames. However, correcting for these violations was more complex than correcting for errors due to bulky side-chain undersampling or salt bridge violations.

For mab2 VH-T → Y, the free energy profile revealed two predominant microstates: one with a distance range of 1.5 Å−2.3 Å and the other with a range of 3.3 Å−5.7 Å. However, there were far too few TI frames in the nearer microstate to separately compute the $\Delta G$ for each and compute a weighted sum weighted by the microstate areas, like with hu4D5−5 VH-W95A. Boosting the sampling in the nearer microstate with NMR restraints was hampered by the bimodal nature of the free energy profile. Therefore, we computed a corrected $\Delta\Delta G$ simply by filtering the TI $\lambda$ production data to the frames conforming to either microstate.

For hu4D5−5 VL-F53N, enforcing NMR restraints for the VL-N53 topology to form an interaction with VL-N53 prevented the VL-F53 topology from properly equilibrating, and the Ab−Ag binding conformation could not accommodate both topologies simultaneously. Thus, we were unable to


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Table 2. Summary of GaMD-Based Corrections to Most Erroneous Predictions[a]

| case | original ΔΔG (kcal/mol) | corrected ΔΔG (kcal/mol) | empirical ΔΔG (kcal/mol) | original error (kcal/mol) | corrected error (kcal/mol) | model $R^2$ | theorized source of error |
|---|---|---|---|---|---|---|---|
| hu4D5−5 W95A | 3.19 ± 0.71 | 4.63 ± 0.47 | 5.07 | 1.88 | 0.44 | 0.53 | bulky side-chain undersampling |
| hu4D5−5 R50A | 2.98 ± 1.02 | 5.14 ± 1.05 | 4.58 | 1.60 | 0.56 | 0.73 | salt bridge violation |
| mab2 VL-Y → R | 0.61 ± 1.04 | −0.87 ± 1.01 | <−0.83 | >1.43 | -[b] | 0.48 | salt bridge violation |
| mab2 VH-T → Y | 1.69 ± 0.75 | 0.61 ± 0.73 | 0 | 1.69 | 0.61 | 0.76 | hydrogen bond violation |
| hu4D5−5 VL-F53N | −0.56 ± 0.57 | - | 1.19 | 1.75 | - | 0.64 | hydrogen bond violation |

[a]For hu4D5-5 R50A and mab2 VL-Y → R, the data shown is for the charging step correction. [b]For mab2 VL-Y → R, although the error reduced meaningfully, the corrected error is not given due to the qualitative empirical data. Uncertainties are 95% confidence intervals determined using bootstrapping.

perform definitive correction for VL-F53N, but the RF model and previous salt bridge results imply that hydrogen bond violation was a significant source of error in this calculation.

**Extending the Error Analysis/Correction Approach to All hu4D5−5 Cases.** To further validate our error analysis and correction approach, we extended the analysis from our 5 most erroneous predictions to the entire hu4D5−5 set of mutations ($n$ = 13). We confirmed that the approach indeed decreased the prediction error in all but 1 case (in which the error increased by 0.44 kcal/mol). The original prediction MAE for the Kelley hu4D5−5 cases was 0.82 ± 0.18 kcal/mol, and after corrections, the MAE decreased to 0.53 ± 0.16 kcal/mol. The RMSE similarly decreased from 1.06 ± 0.22 kcal/mol to 0.70 ± 0.18 kcal/mol (Table 3). For many cases with already accurate predictions, the model did not find any strong correlation between geometric and energetic features in the TI data ($R^2$ < 0.4).

Table 3. Error Analysis of hu4D5-5 ΔΔG Predictions ($n$ = 13) for the Original Protocol,[a] Extended Protocol,[b] and RF + GaMD Correction Approach[c]

| protocol | MAE | RMSE |
|---|---|---|
| original protocol (5 ns per $\lambda$) | 0.82 ± 0.18 | 1.06 ± 0.22 |
| extended protocol (25 ns per $\lambda$) | 0.71 ± 0.18 | 0.93 ± 0.23 |
| RF + GaMD correction | 0.53 ± 0.16 | 0.70 ± 0.18 |

[a]5 ns per $\lambda$ during production. [b]25 ns per $\lambda$ during production. [c]Uncertainties are 95% confidence intervals determined using bootstrapping.

To explore whether similar improvements in accuracy could be achieved simply by using our initial protocol but with longer $\lambda$ production sampling, we re-estimated the ΔΔGs of our entire data set with 25 ns of $\lambda$ production, 5-fold greater than our original 5 ns of $\lambda$ production (Table S7). For the Kelley hu4D5−5 cases, the MAE of these predictions was 0.71 ± 0.18, and the RMSE was 0.93 ± 0.23—only a slight improvement over the original hu4D5−5 predictions (Table 3).

## DISCUSSION

Although in this paper, we analyzed the error in ΔΔ$G$ predictions for canonical amino acid mutations, our motivation for improving the accuracy of these alchemical methods is to eventually enable their application to classifying PTMs during CQA assessment. Decision makers can use alchemical methods to guide CQA control strategy decisions once these methods can reliably predict the ΔΔ$G$ values of perturbations within 0.5 kcal/mol of their empirical values. Currently, our RMSE using cMD-TI and 5 biological replicates is 0.94 ($n$ = 20). This protocol enabled us to correctly estimate the impact of Met oxidation in mab3, which is one of the few attempts in the literature to use TI to estimate the impact of PTMs on Ab−Ag binding affinity. Regardless, because our current protocol still falls short of our accuracy requirements, we closely examined our most inaccurate predictions to identify and mitigate the underlying issues with alchemical methods.

As one of the major findings in our study, we observed key salt bridge violations as a source of error in some of our most inaccurate predictions. The cases hu4D5−5 VH-R50A and mab2 VL-Y → R highlighted the criticality of maintaining key salt bridges during TI, especially those existing at protein−protein interfaces. During charge-changing transformations, the mixed potentials at intermediate $\lambda$ values may weaken the charged residue's ability to form salt bridges and therefore may not accurately account for their contribution to $dV/d\lambda$. This was corroborated by a recent finding by Sampson et al., where around $\frac{1}{3}$ of their erroneous FEP+ predictions of protein−protein complex ΔΔGs were due to disrupted salt bridges.[27] We liken these salt bridge violations to local unbinding events, akin to ligands drifting from the binding pocket during ABFE calculations[95] when protein−ligand interactions are turned off.

ΔΔ$G$ prediction errors arising from such salt bridge violations appear to be a prevalent issue and are not easily remedied with additional sampling. For example, Zhang et al. used alchemical methods to predict the impact of mutations on the binding affinity of barstar-barnase and found that even with up to 100 ns of enhanced sampling via replica exchange between adjacent alchemical states, four out of their twenty-eight ΔΔ$G$ estimations still remained significantly inaccurate.[61] We hypothesized that disrupted salt bridges or hydrogen bonds due to mixed potentials could be a significant source of error in their predictions since this cannot be remedied by enhanced sampling. To explore this hypothesis, we created 1-D free energy profiles derived from GaMD simulations of the wild-type barstar-barnase complex, which revealed salt bridges or hydrogen bonds involving the softcore region for each of these four cases (Figures S5−S11).

Importantly, we found that accurate ΔΔ$G$ predictions in cases with salt bridge violations could be recovered either by filtering for frames containing the intact salt bridge or by enforcing the salt bridge integrity during TI through distance-based restraints. The latter correction method parallels the use of restraints to keep a ligand in place during ABFE calculations as ligand interactions are turned off.[96] ABFE restraints prevent complete unbinding of ligand and cognate protein targets, while our restraints prevent local unbinding (manifesting as disruption of a salt bridge) of an antibody and its cognate antigen.

Notably, Sampson et al. also applied a correction scheme to improve $\Delta\Delta G$ predictions in simulations involving salt bridge violations, but their approach differed substantially from ours. In their study, they devised a custom script to flag potential salt bridge violations and universally applied an empirical correction of $\pm$2.8 kcal/mol based on trends from their large data set.[27] While the Sampson et al. correction is computationally expedient and appropriate for high-throughput use cases, our correction methods (especially the distance restraint to enforce salt bridges) more rigorously address this root cause of the error and are likely more appropriate where throughput demand is low and accuracy is paramount. Future work is necessary to determine how to use restraints to preserve more transient interactions like hydrogen bonds, which may have multimodal end-state free energy profiles.

Note that a fully rigorous implementation of our distance restraint correction would necessitate estimating the free energy contribution due to these restraints. In protein−ligand ABFE calculations, Boresch restraints are used to restrain a noninteracting ligand, which allows the free energy contribution of the restraints to be derived analytically.[95,97] However, such a definitive procedure for protein−protein RBFEs remains an open research question. Given this complexity, we decided to leave this investigation for future work investigating how to account for distance-based restraints in protein−protein RBFE calculations.

Another major finding in our study is that potential sources of error (i.e., improperly sampled yet energetically important DOF) can be identified from simulation data alone in an untargeted search by leveraging machine learning. Previously, Pearson correlation coefficients have been used to identify slow DOF coupled to TI $dV/d\lambda$.[61,98] We used RF models to pinpoint the most energetically relevant DOF because we observed the data to be nonlinear and wanted to account for synergistic effects between multiple DOF. When we applied our RF model to our highest error cases, we consistently found a specific degree of freedom being incorrectly sampled during TI $\lambda$ production when comparing the DOF sampling during TI with an end-state GaMD free energy profile of the DOF. After correcting for the discrepancies we found between TI and end-state GaMD for the most energetically relevant DOF, we observed that the results became more accurate (Table 2), suggesting that our RF + GaMD scheme correctly identified sources of error in our most inaccurate simulations. Applied to the set of 13 hu4D5−5 perturbations, the RF model and GaMD-based correction reduced the RMSE from 1.06 $\pm$ 0.22 to 0.70 $\pm$ 0.18 kcal/mol and the MAE from 0.82 $\pm$ 0.18 to 0.53 $\pm$ 0.16 kcal/mol (Table 3). This set contained a variety of both accurate and inaccurate predictions; thus, this result suggested that the approach is robust enough to improve the inaccurate predictions but not increase the error in the accurate estimations. The RMSE and MAE after corrections (0.70 and 0.53 kcal/mol, respectively) suggest that there is very little error (~0.2 to 0.4 kcal/mol) contributed by the ff14SB[78] force field since Kelley and O'Connell estimated the standard error of the empirical SPR data to be $\pm$0.3 kcal/mol.[70] While our results are quite encouraging, a rigorous analysis of the force-field-associated error in our predictions is required to prove that the RF + GaMD correction definitively improves prediction accuracy.

When examining the erroneous $\Delta\Delta G$ prediction for hu4D5−5 VH-W95A with our approach, we observed that the VH-W95 side-chain rotamers were incorrectly sampled during TI $\lambda$ production. Bulky side-chain undersampling has been established as a source of error in alchemical free energy predictions by many others in the field,[25,62,99−102] which gave us confidence in our RF analysis since the model identified a known source of error in alchemical free energy calculations without being specifically trained to do so. We decided to naively correct for the error in hu4D5−5 VH-W95A using the proportions of the dominant states given by the end-state GaMD free energy profile for VH-W95 chi1/chi2. Although more sophisticated algorithms such as alchemically enhanced sampling (ACES)[62] have been previously published to tackle bulky side-chain undersampling errors, our objective was to verify that our approach comparing TI distributions with GaMD end-state free energy profiles was suitable. After our naive correction was applied, the error in our $\Delta\Delta G$ prediction for hu4D5−5 VH-W95A decreased from 1.88 to 0.44 kcal/mol.

In general, we recommend using our RF model in situations with scant prior knowledge of the DOF essential to the binding interaction. In cases when ACES is needed but the necessary DOF to target are unknown, the RF model can determine which DOF should be included in the enhanced sampling region using only an initial cMD-TI calculation.

## ■ CONCLUSIONS

An *in silico* tool capable of estimating the impact of PTMs on Ab−Ag binding affinity within 0.5 kcal/mol would provide an alternative tool to inform control strategies for potential CQAs during therapeutic biologic development without the laborious enrichment and empirical testing of minor variants. Although alchemical methods are the state of the art for computing binding affinity RBFEs, using alchemical free energy methods in this context still requires a nontrivial improvement in their accuracy. Therefore, we created a data set of alchemical free energy predictions of various Ab−Ag RBFEs and subsequently investigated the sources of error in the most inaccurate predictions.

Our error analysis introduced a novel method which used RF regression to find any nearby DOF (i.e., side-chain rotamers and interatomic distances) with a strong influence on $dV/d\lambda$ and then GaMD to generate an end-state free energy profile of these most influential DOF. When comparing the GaMD free energy profile to our $\lambda$ production sampling in each of our 5 most inaccurate cases, we found improper sampling in highly influential DOF—either incorrect sampling of a bulky side chain or the violation of a key salt bridge or hydrogen bond. These sources of error are highly relevant to estimating the impact of PTMs on bulky or charged residues such as tryptophan oxidation or asparagine deamidation, respectively.

When applied to the set of hu4D5−5 perturbations, our RF + GaMD error analysis approach improved the RMSE of the predictions from 1.06 $\pm$ 0.22 to 0.70 $\pm$ 0.18 kcal/mol. Our model's a priori identification of bulky side-chain undersampling—a known source of error in alchemical free energy calculations—further validated our approach. We corrected for the bulky side-chain undersampling in hu4D5−5 VH-W95A by recomputing the $\Delta G$ of the microstates separately and linearly


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combining them based on GaMD-derived microstate areas. In cases with disrupted salt bridges, we enforced the integrity of the salt bridges using NMR distance restraints. Both correction methods reduced the $\Delta\Delta G$ estimation error by >1 kcal/mol. However, we were not able to find straightforward methods to correct cases in which our model found a disrupted hydrogen bond to be the source of error. We will continue to explore robust methods to enforce convergence of TI sampling to the GaMD free energy profiles.

Ultimately, the main purpose of this work is to motivate the application of alchemical free energy methods to CQA analysis. After validating our protocol with canonical amino acids, we used prediction of the impact of methionine oxidation in mab3, one of the few PTM $\Delta\Delta G$ RBFE predictions in the literature. Moving forward, we will extend our protocol to additional PTMs and compare the impact of various cutting-edge force field parameterization methods on accuracy.

## ■ ASSOCIATED CONTENT

### Data Availability Statement

Initial 5 ns TI data with nearby DOF, code for the RF+GaMD error analysis approach, results from random forest + GaMD error analysis applied to all 13 hu4D5−5 systems and results from extending TI protocol to 25 ns/$\lambda$ are all contained in the following GitHub repository: https://github.com/adnaksskanda/gamdti-paper.

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.5c01135.

> Full empirical data set, full TI predictions data set, results from one-step corrections, explanation of key features referenced in the main text, barstar-barnase GaMD 1-D free energy profiles indicating possible salt bridges or hydrogen bonds (referenced in the Discussion section) (PDF)

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### Author Contributions

M.K. conceived the project, designed the study, and curated the experimental benchmarking data. S.S. performed the MD simulations, conceived the juxtaposition of TI with GaMD, designed the random forest model, and generated the corrected TI predictions. S.S. and M.K. cowrote the manuscript, with M.K. providing the high-level framework and S.S. generating the figures and first drafts.


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### Notes

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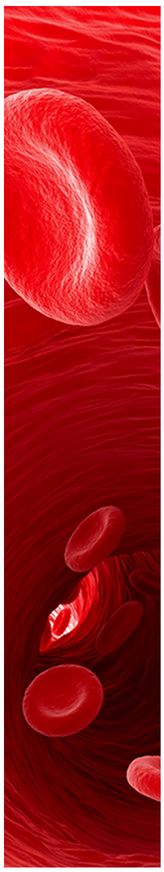





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